SMART POLE

MONITORING AND CONTROL SYSTEM

1. Project Overview

The recommended project is a Smart Pole Monitoring and Control System. The goal is not simply to make a lamp turn ON and OFF. The interns should build a small but complete IoT/embedded system demonstrating hardware control, sensor measurements, communication, a central monitoring system, software engineering practices, and technical documentation.

Project Objective

Design and develop a functional smart pole prototype consisting of a pole-side controller and a centralized management system. The pole controller controls electrical devices, collects sensor readings, and communicates with the central system. The central system allows operators to monitor the pole, view electrical measurements, and remotely control its functionalities.

2. What the System Should Do

- Lamp control: Turn the streetlight ON/OFF remotely through the central system.
- Light dimming: Adjust lamp brightness using a command from the central system (optional extension).
- LCD display: Display messages sent from the central system, including individual letters or text.
- Temperature monitoring: Measure temperature inside or around the pole's electrical enclosure.
- Electrical and Power monitoring: Measure voltage and current for each monitored electrical output.
- Central communication: Transmit sensor readings and device status to the central system.

3. System Architecture

The Team should implement the following high-level architecture:

Layer	Main elements	Responsibilities
Central Management System	Web dashboard, API, database, command management	Display live readings and history; allow authorized operators to issue commands; store telemetry and events.
Communication Layer	MQTT and/or HTTP/REST API; JSON payloads..etc	Carry telemetry from the pole to the central system and commands from the central system to the pole.
Pole Controller	suitable embedded controller	Read sensors, control outputs, validate commands, manage connectivity, and report status.
Field Devices and Sensors	Lamp, LCD, temperature sensor, voltage sensors, current sensors	Provide physical outputs and measurements to/from the pole controller.

4. Functional Requirements and Priority Levels

Mandatory requirements must be completed. Optional requirements are extensions that may be attempted after the mandatory scope is working.

Requirement	Description	Priority
Simulation for Prototype	Simulation on proteus, multisim,,,	MANDATORY
Lamp ON/OFF	Remotely switch the lamp ON and OFF.	MANDATORY

LCD control	Send text or individual letters from the central system to the LCD.	MANDATORY
Temperature	Measure and transmit pole/enclosure temperature.	MANDATORY
Voltage measurement	Measure voltage for each monitored electrical output.	MANDATORY
Current measurement	Measure current for each monitored electrical output.	MANDATORY
Power calculation	Calculate power consumption for each monitored output.	MANDATORY
Telemetry	Send readings and device status to the central system.	MANDATORY
Central dashboard	Display live readings and pole status.	MANDATORY
Remote dimming	Adjust lamp brightness through the central system.	OPTIONAL
Automatic dimming	Automatically adjust brightness according to ambient light or a schedule.	OPTIONAL
Fault detection	Detect abnormal readings and generate alerts.	OPTIONAL
Historical monitoring	Store and display historical sensor readings.	OPTIONAL
Automatic scheduling	Configure scheduled lamp ON/OFF operations.	OPTIONAL

4. Division of Work

Primary module while requiring all members to understand the complete system.

Assigned module	Main responsibilities
Embedded system	Controller setup, hardware architecture, firmware structure, device communication.
Electrical monitoring	Voltage, current, power measurement, temperature measurement, and calibration.
Lamp and LCD control	Lamp switching, LCD communication, remote display messages, and optional dimming.
Backend and communication	MQTT broker, REST API, telemetry processing, command handling, and database.
Frontend and integration	Dashboard, live readings, controls, testing, integration, and system demonstration.

5. Software Engineering and Coding Standards

The project must not be implemented as one large source file. The team must follow professional software engineering practices.

Area	Requirement	Priority
Code structure	Separate modules, functions, and responsibilities.	MANDATORY
Naming conventions	Use consistent variable, function, and file naming.	MANDATORY
Error handling	Handle sensor errors, invalid commands, and communication failures.	MANDATORY

Version control	Use Git and GitHub.	MANDATORY
Logging	Record errors, events, and important operations.	MANDATORY
Testing	Perform unit tests and integration tests.	MANDATORY
API documentation	Document every API endpoint, request, response, and error.	MANDATORY
README	Explain installation, configuration, and execution.	MANDATORY
Architecture documentation	Document the complete system architecture.	MANDATORY

6. Project Development Phases

Phase	Activities	Expected output
Phase 1	Research, requirements, and architecture.	Approved system design.
Phase 2	Component selection and hardware development.	Working hardware prototype.
Phase 3	Backend and communication.	Controller communicating with the central system.
Phase 4	Dashboard development.	Working monitoring and control dashboard.
Phase 5	Integration and testing.	Complete smart pole prototype.
Phase 6	Documentation and presentation.	Final report, GitHub repository, and demonstration.